1. Suppose that the distribution function of X is given by

$$F(b) = \begin{cases} 0 & b < 0 \\ \frac{b}{4} & 0 \le b < 1 \\ \frac{1}{2} + \frac{b-1}{4} & 1 \le b < 2 \\ \frac{11}{12} & 2 \le b < 3 \\ 1 & 3 \le b \end{cases}$$

a) Find $P\{\frac{1}{2} < X < \frac{3}{2}\}$.

b) Find $P\{X = i\}, i = 1, 2, 3$.

2. An image is partitioned into two regions, one white and the other black. A reading taken from a randomly chosen point in the white section will be normally distributed with $\mu = 4$ and $\sigma^2 = 4$, whereas one taken from a randomly chosen point in the black region will have a normally distributed reading with parameters $(6,9)$.

A point is randomly chosen on the image and has a reading of 5.

If the fraction of the image that is black is α, for what value of α would the probability of making an error be the same, regardless of whether one concluded that the point was in the black region or in the white region?

3. Three points X_1, X_2, X_3 are selected at random on a line L. What is the probability that X_2 lies between X_1 and X_3?

4. Let X be a normal random variable with mean 12 and variance 4.

a) Find the value k such that $P\{X > k\} = 0.10$,

b) Determine $P\{X < 10\}$.

5. Each item produced by a certain manufacturer is, independently, of acceptable quality with probability 0.95. Approximate the probability using normal distribution that at most 10 of the next 150 items produced are unacceptable.

6. E-mails arrive one at a time in an inbox. The waiting times between e-mails are independent and identically distributed with $Exponential(\lambda)$.

Each e-mail is non-spam with probability p, and spam with $q = 1 - p$ (independently of other e-mails and of the waiting times.

Let X be the time at which the first non-spam e-mail arrives. Find

a) $E[X]$,

b) $Var(X)$.

7. The monthly worldwide average number of airplane crashes of commercial airlines is 3.5. What is the expected amount of time (in months) until the next airplane crash, starting from today?

8. A post office has two clerks, where the first clerk has a service time with Expo(λ_1) distribution, and the second clerk has service time with Expo(λ_2) distribution. When you enter the post office, you see both clerks busy (i.e., serving a customer). If you are next in line, what is the expected amount of time that you will need to wait for a clerk to become available to serve you?

9. If bolt thread length is normally distributed, what is the probability that the thread length of a randomly selected bolt is

a) Within 1.5 standard deviations of its mean value?

b) Between 1 and 2 standard deviations above its mean value?

10. A stick is broken into two pieces, at a uniformly chosen break point. For the length of the longer piece, find

a) the cdf;

b) mean value.

11. The following table uses 2006 data concerning the percentages of male and female full-time workers whose annual salaries fall into different ranges:

Earnings Range	Percentage of Females	Percentage of Males
≤9999	8.6	4.4
10,000 – 19,999	38.0	21.1
20,000 – 24,999	19.4	15.8
25,000 – 49,999	29.2	41.5
≥50,000	4.8	17.2

Suppose that random samples of 200 male and 200 female full-time workers are chosen. **Use Normal approximation** to find the probability that

a) at least 70 of the women earn $25,000 or more;

b) at most 60% of the men earn $25,000 or more;

c) at least 75% of the men and at least half the women earn $20,000 or more.

12. There are two machines available for cutting corks intended for use in wine bottles:

i. The first machine produces corks with diameters that are normally distributed with mean 3 cm and standard deviation 0.1 cm.

ii. The second machine produces corks with diameters that have a normal distribution with mean 3.04 cm and standard deviation .02 cm.

If acceptable corks have diameters between 2.9 cm and 3.1 cm;

a) Which machine is more likely to produce an acceptable cork?

b) If machine 1 produces 60% of all corks, and a randomly selected cork is found to be acceptable, then what is the probability that this cork was produced by the first machine?

SOLUTIONS

1. Suppose that the distribution function of X is given by

$$F(b) = \begin{cases} 0 & b < 0 \\ \frac{b}{4} & 0 \le b < 1 \\ \frac{1}{2} + \frac{b-1}{4} & 1 \le b < 2 \\ \frac{11}{12} & 2 \le b < 3 \\ 1 & 3 \le b \end{cases}$$

a) Find $P\{\frac{1}{2} < X < \frac{3}{2}\}$.

$$P\left\{\frac{1}{2} < X < \frac{3}{2}\right\} = P\left\{X < \frac{3}{2}\right\} - P\left\{X < \frac{1}{2}\right\} = F\left(\frac{3}{2}\right) - F\left(\frac{1}{2}\right) = \left(\frac{1}{2} + \frac{\left(\frac{3}{2}\right) - 1}{4}\right) - \frac{\left(\frac{1}{2}\right)}{4} = \left(\frac{1}{2} + \frac{1}{8}\right) - \frac{1}{8} = \frac{1}{2}$$

b) Find $P\{X = i\}, i = 1, 2, 3$.

$$P\{X = i\} = F(i) - F(i-1)$$

$$P\{X = 1\} = F(1) - F(0) = \left(\frac{1}{2} + \frac{1-1}{4}\right) - 0 = \frac{1}{2}$$

$$P\{X = 2\} = F(2) - F(1) = \frac{11}{12} - \left(\frac{1}{2} + \frac{1-1}{4}\right) = \frac{11}{12} - \frac{6}{12} = \frac{5}{12}$$

$$P\{X = 3\} = F(3) - F(2) = 1 - \frac{11}{12} = \frac{1}{12}$$

2. An image is partitioned into two regions, one white and the other black. A reading taken from a randomly chosen point in the white section will be normally distributed with $\mu = 4$ and $\sigma^2 = 4$, whereas one taken from a randomly chosen point in the black region will have a normally distributed reading with parameters $(6,9)$.
A point is randomly chosen on the image and has a reading of 5.
If the fraction of the image that is black is α, for what value of α would the probability of making an error be the same, regardless of whether one concluded that the point was in the black region or in the white region?

Let W denote a random point taken from the white region, and B a random point from the black region.

If both regions were of the same size, the probability of making an error would be the same when:
$P(W > 5) = P(B < 5)$

Since the black region is of size α, the white region would be of size $1 - \alpha$. Hence, the odds for picking a random point from the black region against the white region is "α: $(1 - \alpha)$". So, the probability of making an error would be the same when:

$$(1 - \alpha)P(W > 5) = P(B < 5)\ \alpha$$

$$(1 - \alpha)P\left(z_W > \frac{5-4}{2}\right) = P\left(z_B < \frac{5-6}{3}\right)\ \alpha$$

$$(1 - \alpha)P(z_W > 0.5) = P(z_B < -0.33)\ \alpha$$

$$(1 - \alpha)(1 - 0.69146) = (0.37070)\ \alpha$$

$$(1 - \alpha)(0.30854) = (0.37070)\ \alpha$$

$$0.30854 = (0.37070 + 0.30854)\ \alpha$$

$$0.30854 = (0.67924)\ \alpha$$

$$\frac{0.30854}{0.67924} = \alpha$$

$$0.45424 \cong \alpha$$

3. Three points X_1, X_2, X_3 are selected at random on a line L. What is the probability that X_2 lies between X_1 and X_3?

Since X_1, X_2, X_3 are all selected randomly from the same line, they have the same distribution (i.e., the probabilities associated with them are the same). → *This explanation is necessary for the following calculations:*

Because of the above statement, the probabilities that any of these 3 points to be in the middle are the same. There are two different outcomes where X_2 is in the middle: $X_1 < X_2 < X_3$ and $X_3 < X_2 < X_2$. There are 6 different outcomes. Hence the probability that X_2 is in the middle is:

$$\frac{2}{6} = \frac{1}{3}$$

An alternative way to solve this problem is to define $X_i \sim Uniform(0, L)$ and calculate the probability in interest, which will also yield the solution 1/3.

4. Let X be a normal random variable with mean 12 and variance 4.

a) Find the value k such that $P\{X > k\} = 0.10$,

$$P(X > k) = P\left(\frac{X-\mu}{\sigma} > \frac{k-12}{2}\right) = P\left(Z > \frac{k-12}{2}\right) = 0.10$$

$$z_{0.10} = 1.28 = \frac{k-12}{2}$$

$$k = (2)(1.28) + 12 = 14.56$$

b) Determine $P\{X < 10\}$.

$$P(X < 10) = P\left(Z < \frac{10-12}{2}\right) = P(Z < -1) = 0.15866$$

5. Each item produced by a certain manufacturer is, independently, of acceptable quality with probability 0.95. Approximate the probability using normal distribution that at most 10 of the next 150 items produced are unacceptable.

$$\mu = np = (150)(0.95) = 142.5$$

$$\sigma^2 = np(1-p) = (142.5)(0.05) = 7.125$$

$$\sigma = \sqrt{7.125} \cong 2.67$$

At most 10 unacceptable units implies at least 140 acceptable units:

$$P(X \geq 140) = P\left(Z > \frac{139.5 - 142.5}{2.67}\right) \cong P(Z > -1.12) = P(Z < 1.12) \approx 0.86864$$

6. E-mails arrive one at a time in an inbox. The waiting times between e-mails are independent and identically distributed with $Exponential(\lambda)$.

Each e-mail is non-spam with probability p, and spam with $q = 1 - p$ (independently of other e-mails and of the waiting times.

Let X be the time at which the first non-spam e-mail arrives. Find

a. $E[X]$,

b. $Var(X)$.

Note that the arrival of the first non-spam e-mail is a geometric random variable, with mean $\frac{1}{1-p}$.

Using the linearity of the E[.] function:

$$E[X] = \left(\frac{1}{1-p}\right)\left(\frac{1}{\lambda}\right)$$

Hence,

$$Var(X) = \left(\frac{1}{1-p}\right) \left(\frac{1}{\lambda}\right)^2$$

7. The monthly worldwide average number of airplane crashes of commercial airlines is 3.5. What is the expected amount of time (in months) until the next airplane crash, starting from today?

Let X = "time until the next crash"

The airplane crashed can be modeled with the Poisson process with $\lambda = 3.5$.

Hence, time in-between two crashes is an Exponential distribution with mean $\frac{1}{\lambda} = \frac{1}{3.5} = \frac{2}{7}$ months.

Due to the memoryless property of the exponential distribution, the answer is again 2/7 months.

8. A post office has two clerks, where the first clerk has a service time with Expo(λ_1) distribution, and the second clerk has service time with Expo(λ_2) distribution. When you enter the post office, you see both clerks busy (i.e., serving a customer). If you are next in line, what is the expected amount of time that you will need to wait for a clerk to become available to serve you?

$X = \min(X_1, X_2)$

$P(X > x) = P(\min(X_1, X_2) > x) = P(X_1 > x, X_2 > x) = P(X_1 > x)P(X_2 > x)$

$= e^{-\lambda_1 x} e^{-\lambda_2 x} = e^{-(\lambda_1+\lambda_2)x}$

$X \sim Expo(\lambda_1 + \lambda_2)$

$E[X] = \dfrac{1}{\lambda_1 + \lambda_2}$

9. If bolt thread length is normally distributed, what is the probability that the thread length of a randomly selected bolt is

a) Within 1.5 standard deviations of its mean value?

$P(\mu - 1.5\sigma \le X \le \mu + 1.5\sigma) = P(-1.5 \le Z \le 1.5)$

$= \Phi(1.5) - \Phi(-1.5) = 0.93319 - 0.06681 = 0.86638$

b) Between 1 and 2 standard deviations <u>above</u> its mean value?

$P(\mu + \sigma \le X \le \mu + 2\sigma) = P(1 \le Z \le 2)$

$= \Phi(2) - \Phi(1) = 0.97725 - 0.84134 = 0.13591$

10. A stick is broken into two pieces, at a uniformly chosen break point. For the length of the longer piece, find

We can safely assume that the stick is of length 1. Then let L = "length of the longer piece" and U = "break point on stick". Then

$$L \sim Unif\left(\frac{1}{2}, 1\right)$$

$$U = Unif(0,1)$$

a) the cdf;

$$F_L(x) = P(L < x) = P(1 - x < U < x) = P(U < x) - P(U < 1 - x) = x - (1 - x) = 2x - 1$$

b) mean value.

$$E[L] = \frac{1 + 1/2}{2} = 3/4$$

or the midpoint of the interval $\left(\frac{1}{2}, 1\right)$.

11. The following table uses 2006 data concerning the percentages of male and female full-time workers whose annual salaries fall into different ranges:

Earnings Range	Percentage of Females	Percentage of Males
≤9999	8.6	4.4
10,000 – 19,999	38.0	21.1
20,000 – 24,999	19.4	15.8
25,000 – 49,999	29.2	41.5
≥50,000	4.8	17.2

Suppose that random samples of 200 male and 200 female full-time workers are chosen. Use Normal approximation to find the probability that

a) at least 70 of the women earn $25,000 or more;
b) at most 60% of the men earn $25,000 or more;
c) at least 75% of the men and at least half the women earn $20,000 or more.

Let M_i and W_i be the number of men and women, respectively, who earn at least \$ik per year.

a) $P(W_{25} \geq 70) \cong P(W_{25} \geq 69.5)$

$$= P\left(Z_{25} \geq \frac{69.5 - (200)(0.34)}{\sqrt{200(0.34)(0.66)}}\right) = P(Z \geq 0.2239) = 0.41294\ (0.41683)$$

b) $P(M_{25} \leq 120) \cong P(M_{25} \leq 120.5)$

$$= P(Z_{25} \leq \frac{120.5 - (200)(0.587)}{\sqrt{200(0.587)(0.413)}} = P(Z \leq 0.4452) = 0.67003\ (0.67364)$$

c) $P(W_{20} \geq 100) \cong P(W_{20} \geq 99.5)$

$$= P\left(Z_{20} \geq \frac{99.5 - (200)(0.534)}{\sqrt{200(0.534)(0.466)}}\right) = P(Z \geq -1.0348) = 0.84849\ (0.85083)$$

$P(M_{20} \geq 150) \cong P(M_{20} \geq 149.5)$

$$= P\left(Z_{20} \geq \frac{149.5 - (200)(0.745)}{\sqrt{200(0.745)(0.255)}}\right) = P(Z \geq 0.0811) = 0.46812\ (0.46414)$$

Since they are independent:

$$P(W_{20} \geq 100)P(M_{20} \geq 150) = (0.84849)(0.46812) \approx 0.3972$$

12. There are two machines available for cutting corks intended for use in wine bottles:

iii. The first machine produces corks with diameters that are normally distributed with mean 3 cm and standard deviation 0.1 cm.

iv. The second machine produces corks with diameters that have a normal distribution with mean 3.04 cm and standard deviation .02 cm.

If acceptable corks have diameters between 2.9 cm and 3.1 cm;

a) Which machine is more likely to produce an acceptable cork?

Let X_i : the diameter of a randomly selected cork made by the ith machine

$$P(2.9 \le X_1 \le 3.1) = P(\frac{2.9-3.0}{0.1} \le \frac{X_1-\mu_1}{\sigma_1} \le \frac{3.1-3.0}{0.1}) = P(-1.00 \le Z_1 \le 1.00) = 0.6826$$

$$P(2.9 \le X_2 \le 3.1) = P\left(\frac{2.9-3.04}{0.02} \le \frac{X_2-\mu_2}{\sigma_2} \le \frac{3.1-3.04}{0.02}\right) = P(-7.00 \le Z \le 3.00) = 0.9987$$

So, the second machine wins handily.

b) If machine 1 produces 60% of all corks, and a randomly selected cork is found to be acceptable, then what is the probability that this cork was produced by the first machine?

Let *A* = "the cork is acceptable" and *B* = "the first machine was used." Then

$$P(B|A) = \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|B')P(B')} = \frac{(0.6826)(0.6)}{(0.6826)(0.6) + (0.4)(0.9987)} = 0.5062$$